

Eric Zabala

Software Developer Full Stack

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Skills

- JavaScript, Python, Java, C#
- DOM Manipulation
- REST API Development
- .NET (C#), Flask, React.js
- MySQL, SQL Server
- AWS, Cloudflare
- Linux, Windows, macOS
- Object-Oriented Programming
- Data Structures & Algorithms
- Effective Communication
- Troubleshooting
- Team-Oriented Collaboration

Professional Experience

Personal Shopper: Walmart

May 2019 – Dec 2019, April 2021-Aug 2022

- Selected and quality-checked items for customers during online grocery orders.
- Maintained accurate inventory levels, restocking shelves efficiently, and visual merchandising.

DNS Technician: Dish Network

Dec 2019 - March 2021

- Installed satellite dishes and troubleshooted cable issues for customers during their time of need.
- Guided customers in selecting the ideal smart home solutions tailored to their needs.

Senior Radar Operator: Army

Nov 2015 – May 2019

- Managed a small team promoting moral and physical health while performing daily duties.
- Facilitated the maintenance of 18 million dollars worth of highly sensitive equipment.
- Trained soldiers on AN/TPQ-53 radar system operations, map reading, vehicle PCMS, and NVG operation.

Education

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Aug 2021 – Dec 2025

Marshall University

Huntington, West Virginia

Graduating with honors, Nominee for the Maier Award, Dean's list: fall 2022 - spring 2024, SVA Officer,

Published a Research project in an IEEE Journal

Projects

AI-based Education System – Full Stack Web Application & Research

Jun 2024-Aug 2024

- **Frontend:** Developed the quiz flow, exam submissions, chat UI, and feedback display.
- **Backend Infrastructure:** Built a multi-route Flash backend with REST endpoints.
- **Hosting of the website:** Managed the deployment of the website on AWS using EC2.
- **AI-Integrated services:** Connected backend logic to OpenAI API for exam grading, real-time Q&A, quiz grading and personalized course recommendation
- **Research Contribution:** Co-authored a peer-reviewed paper presented at **IEEE UEMCON 2024** on the system's design and evaluation of system capabilities.

Parking spot detection system - Full Stack Web Application

Aug 2025 – Present

- **Frontend:** Built the UI, admin login, live camera feed display, real-time slot status, lot controls.
- **Backend:** Implemented a Flash-based REST API providing endpoints for lots, sots, polygons, plain-slot layouts, and live video streaming from camera
- **Database Layer:** Designed MYSQL models with relations. Constraints, and unique indexes for support CURE operations and cascade deletes.
- **System Integration:** Connected the Flask API, detection service, and JavaScript frontend to deliver real-time updates across the system.

Healthcare System – Full Stack Web Application

Jan 2025 – May 2025

- **Authentication and Session Management:** Implemented secure login, logout, and password recovery with role validation and session handling
- **Messaging and prescriptions:** Built backend CRUD functionality for message storing and prescriptions tracking, with tailored frontend pages for each user role.
- **UI Components:** Built a shared, modular sidebar with dynamically rendered content based upon user role and updated across all pages automatically.
- **Data layer:** Initially built using SQL Server and SSMS, later migrated to MySQL. Integrated with a .NET API.

Memory Game – React Project

Dec 2025 – Present

- **Component-Based Architecture:** Built the game using React functional components with a clear separation between components
- **State Management:** Managed states with React hooks (useState) to track card selections, move counts, matched pairs, and win conditions.
- **Conditional Rendering:** Dynamically rendered UI elements based on game state, including move tracking, card flipping, and win screen logic.
- **User Interaction Logic:** Implemented safeguards to prevent invalid actions such as flipping more than two cards or re-selecting matching cards or the same card twice.
- **Deployment & Routing:** Deployed application using Netlify and with redirect configuration to support client-side routing.

Freelance Website Development Studio – React Project

May 2025

- **Custom Contact Form:** Integrated a custom-built contact form using EmailJS to send inquiries directly to my email without a backend server
- **Styling and Components:** Used modular components and custom CSS to create a responsive, mobile friendly layout across the entire application.
- **Deployment:** Hosted the website on Netlify with the use of environment variables for secure form handling and site metadata

Calculator – React Project

Dec 2025

- **Component-Based Architecture:** Built the calculator using React functional components with clear separation between display, keypad and logic handling.
- **Input Validation & Logic:** Implemented operator handling and input validation to prevent invalid expression and runtime errors.
- **Deployment & Routing:** Deployed application using Netlify

Portfolio Website – Frontend Web Application

Jul 2024 - Present

- **SPA Architecture:** Implemented as a single-page application for seamless user experience and dynamic content loading.
- **Technological Stack:** Utilized HTML, CSS, and JavaScript for robust frontend development.
- **Hosting and Version Control:** Hosted on GitHub Pages, demonstrating proficiency in using GitHub for version control and deployment.

Python Hangman Game – Introductory Programming Project

Jan 2022

Programmed a command-line Hangman game in Python as an introductory project to explore fundamental programming concepts. This early work laid the groundwork for subsequent projects in advanced programming languages.

Java Hangman Game with GUI – Object-Oriented Programming Project

Jan 2023

- Designed and implemented an interactive Hangman game in Java, featuring a custom user interface built with JFrame. Incorporated object-oriented programming principles to ensure a robust and user-friendly gaming experience.

Java Bank Application – Data Structures Project

Jan 2023-Dec 2023

- Engineered custom data structures (linked lists, AVL trees, etc.) within a Java bank application to streamline record management, enabling sophisticated sorting, searching, and merging operations via a console interface.